

SALES PIPELINE TRACKER WITH AI INSIGHTS

End-to-End Low-Code CRM Solution
Microsoft Power Platform Portfolio Project

Power Apps • Power Automate • Dataverse • Power BI • Copilot Studio

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1. EXECUTIVE SUMMARY

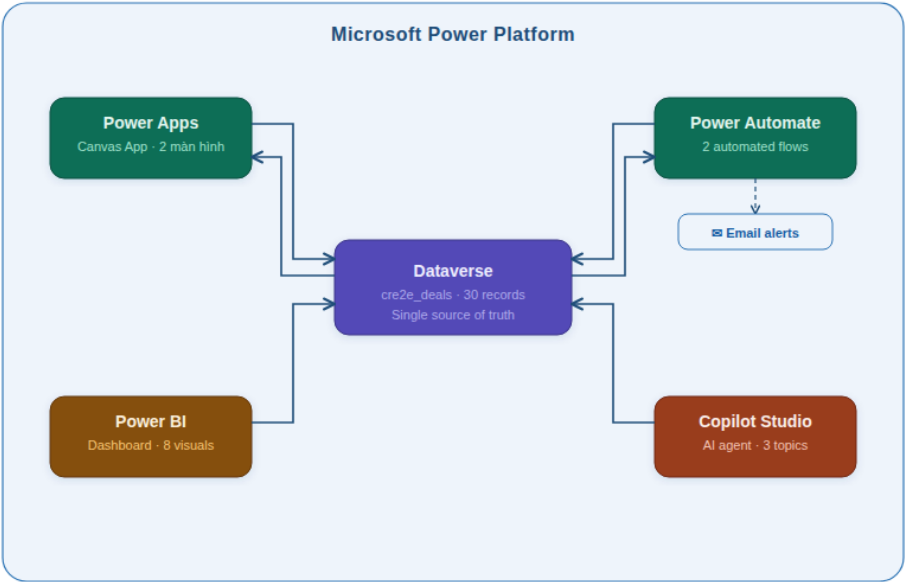
This report presents the complete development process of a Sales Pipeline Tracker system built entirely on the Microsoft Power Platform. The system manages 30 sales deals across 5 pipeline stages and 3 sales representatives, providing real-time data management, process automation, multi-dimensional BI analytics, and an interactive AI assistant.

Key Highlights

- The system is deployed across 5 Power Platform components within a single environment, with all components connected and syncing data through Dataverse.
- Power Apps Canvas App enables users to add/view deals directly, connects live to Dataverse, and displays exactly 30 records to end users.
- 2 Power Automate flows handle all automated email notifications, including conditional logic that sends a congratulation email only when Stage = "Won".
- Power BI dashboard with 8 visual types and 1 custom DAX measure, reflecting the full pipeline from KPI cards to monthly and stage-level analysis.
- Copilot Studio AI agent with 3 topics, implementing conditional branching and choice variables, enabling natural language queries against pipeline data.

2. SYSTEM ARCHITECTURE

The system is designed with a layered architecture, where Dataverse serves as the Single Source of Truth. All components read/write data from a single location, ensuring consistency and scalability.



Overall Architecture

2.1 Architecture Layers

Layer	Component	Role
Data	Dataverse	Central storage - table cre2e_deals with 30 records, 7 columns
Application	Power Apps	Canvas App for deal management (CRUD operations)
Automation	Power Automate	Automated email notifications triggered by Dataverse events
Analytics	Power BI	BI Dashboard with DAX measures and multi-dimensional visuals
AI	Copilot Studio	Natural language chatbot for querying pipeline data

2.2 Deployment Environment

- Platform: Microsoft Power Platform
- Account: UEH Student Account
- Data: 30 synthetic deals, Jan–Apr 2025, 3 sales reps (Minh, Hoa, Tuan)
- All components share 1 environment, connected directly through Dataverse

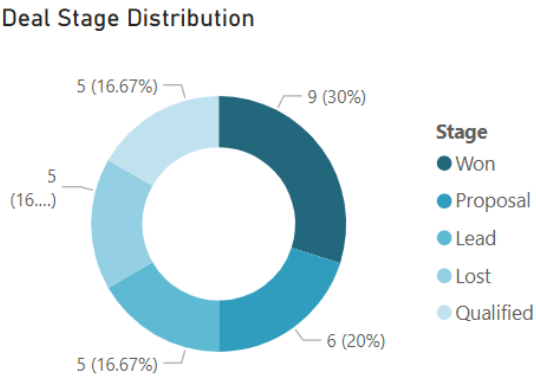
3. DATA FOUNDATION - DATAVERSE

Dataverse was selected as the data foundation due to its native integration with the entire Power Platform, built-in security model, and support for structured data relationships.

3.1 Table Design: cre2e_deals

Column	Data Type	Description
Deal Name	Single line of text	Deal Name (Primary key)
Customer Name	Single line of text	Customer / Account name
Deal Value	Currency	Deal value in USD
Stage	Choice	5 options: Lead / Qualified / Proposal / Won / Lost
Expected Close Date	Date only	Target deal close date
Sales Rep	Single line of text	Responsible rep: Minh, Hoa, or Tuan
Notes	Multiline text	Free-form notes about the deal

3.2 Data Distribution



Deal Distribution by Stage

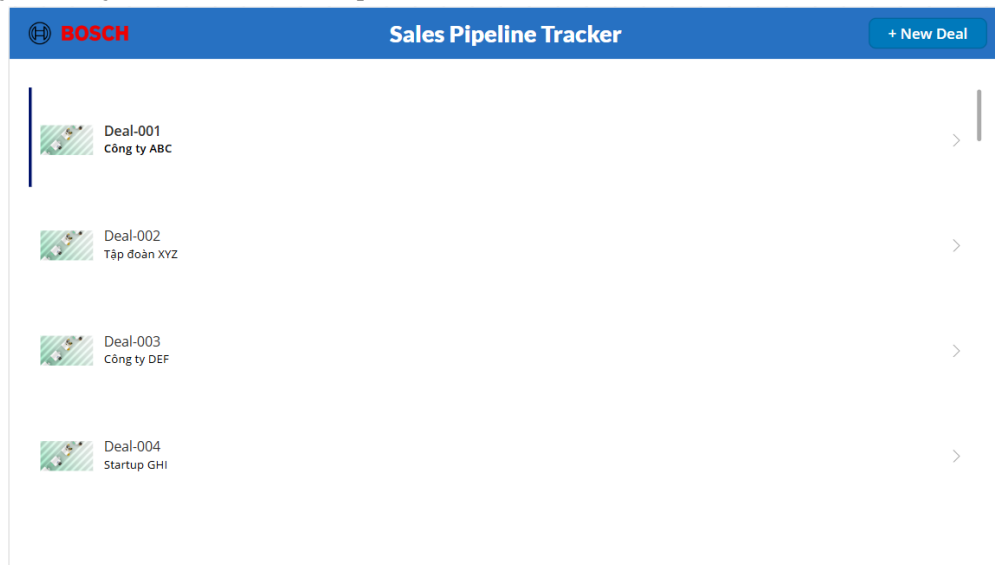
Stage	Deal Count	Percentage
Lead	5	17%
Qualified	5	17%
Proposal	6	20%
Won	9	30%
Lost	5	17%

4. POWER APPS - CANVAS APPLICATION

A Tablet-format Canvas App was built to provide an intuitive deal management interface, allowing sales reps to browse the deal list and add new deals without installing any software.

4.1 Screen 1: DealListScreen

- Header bar with app title
- Vertical Gallery connected directly to the Dataverse table, displaying Deal Name + Customer Name
- “+ New Deal” button navigates to DealFormScreen
- Displays exactly 30 records on startup



DealListScreen

4.2 Screen 2: DealFormScreen

- Edit Form (DealForm) bound to the Dataverse table
- 7 input fields: Deal Name, Customer Name, Deal Value, Stage, Expected Close Date, Sales Rep, Notes
- “Save Deal” button: SubmitForm(DealForm) + Navigate(DealListScreen)
- “← Back” button: Navigate(DealListScreen) without saving

 **BOSCH**

Sales Pipeline Tracker

← Back

* Deal Name

Customer Name

Deal Value

Stage

▼

Expected Close Date

December 31, 20

00

▼

:

00

▼

Sales Rep

Notes

Save Deal

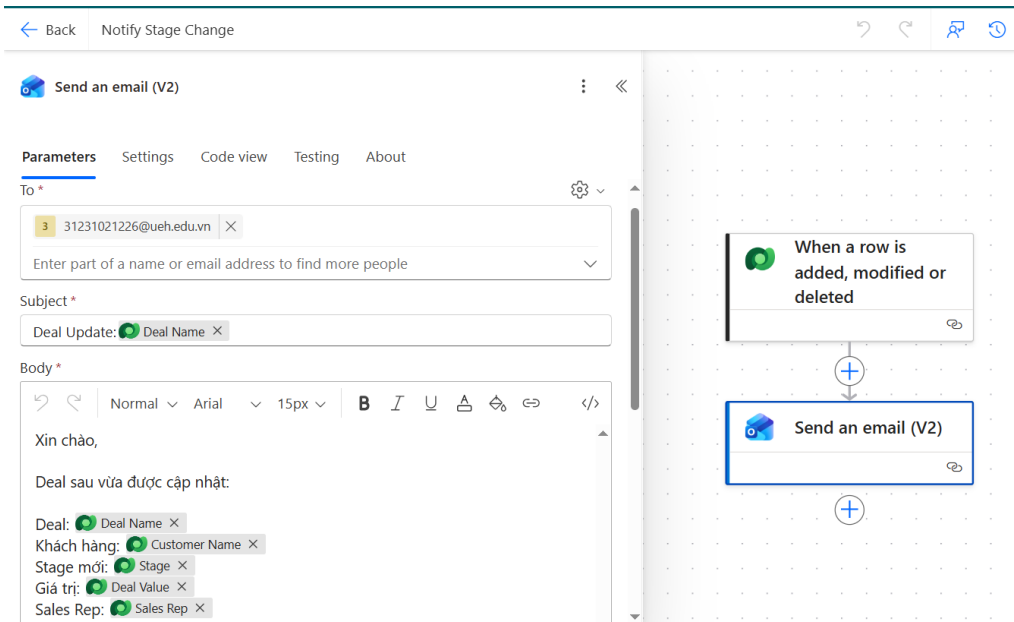
DealFormScreen

5. POWER AUTOMATE - PROCESS AUTOMATION

Two automated cloud flows were built based on Dataverse row-modification events, demonstrating conditional logic and integration with Office 365 Outlook.

5.1 Flow 1: Notify Stage Change

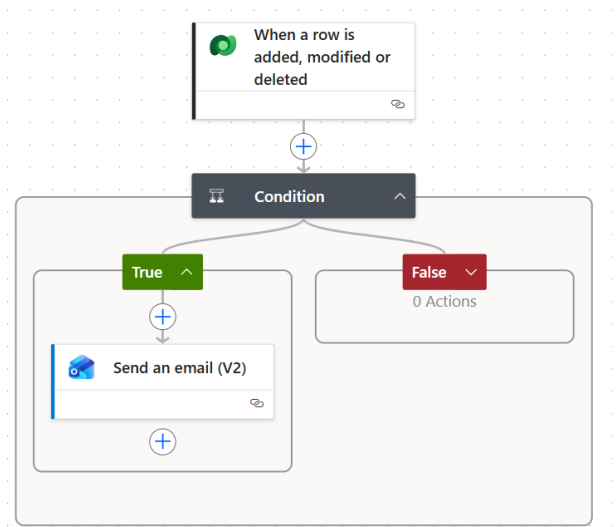
Property	Details
Trigger	When a row is modified (Dataverse - cre2e_deals)
Action	Send an email (V2) qua Office 365 Outlook
Email Content	Deal Name, Customer Name, Stage, Deal Value, Sales Rep
Scope	Triggered when any row is modified



Flow 1 Diagram: Notify Stage Change

5.2 Flow 2: Congrats Won Deal

Property	Details
Trigger	When a row is modified (Dataverse - cre2e_deals)
Condition	Stage equals 'Won'
If Yes	Send congratulation email to Sales Rep
If No	No action taken (terminate branch)



Flow 2 Diagram: Congrats Won Deal

6. POWER BI - ANALYTICS DASHBOARD

The dashboard is connected to Dataverse in Import mode, including a data cleaning step via Power Query and a custom DAX measure, providing a comprehensive overview of pipeline performance.

6.1 Data Preparation (Power Query)

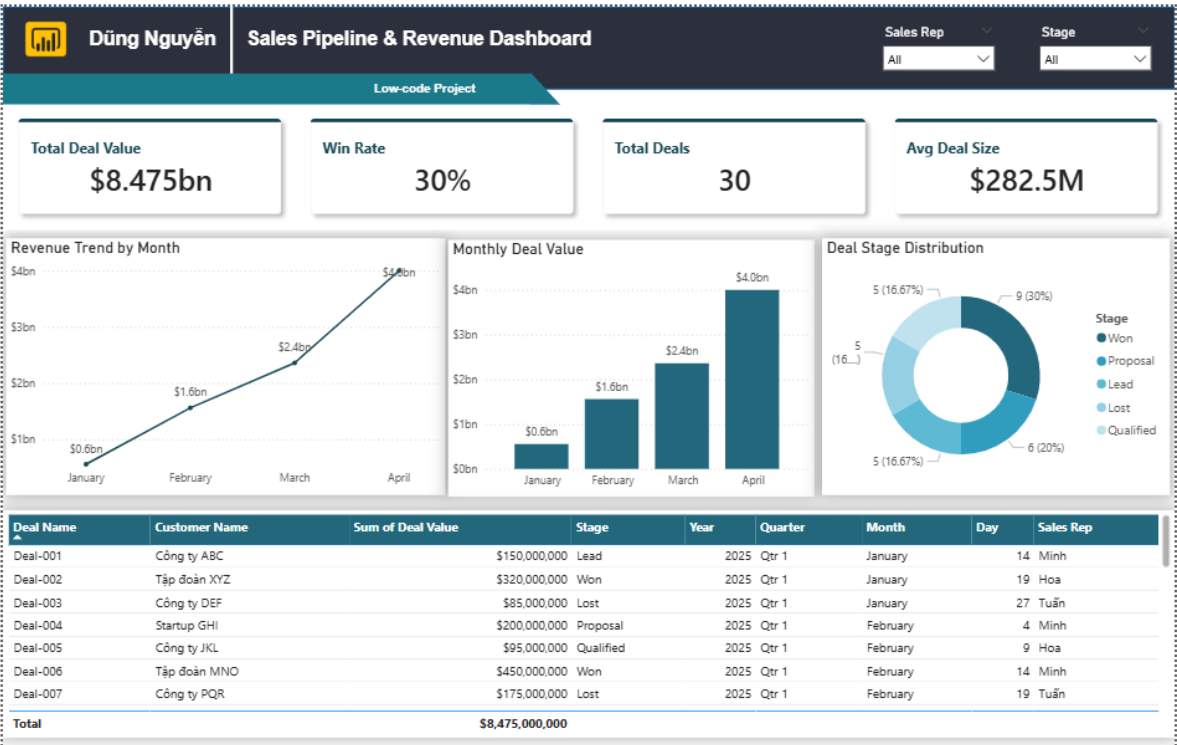
- Remove the cre2e_stage column (numeric), retain cre2e_stagename (text format)
- Rename columns to user-friendly display names for reporting

6.2 DAX Measure

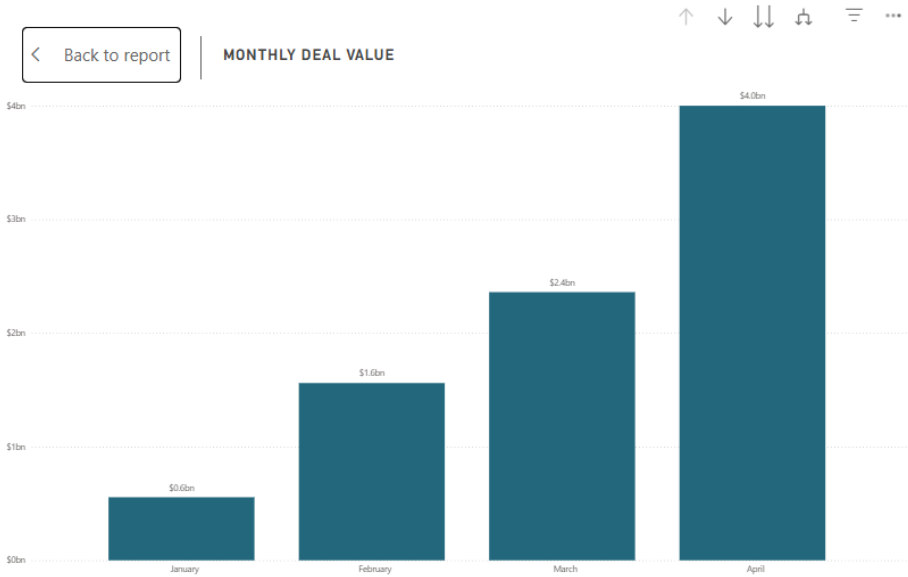
```
Win Rate = DIVIDE(COUNTROWS(FILTER('cre2e_deals', 'cre2e_deals'[Stage] = "Won")),  
COUNTROWS('cre2e_deals'), 0) * 100
```

6.3 Visual List

Visual Type	Title	Content
Card	Total Pipeline Value	\$8.475 Billion (SUM Deal Value)
Card	Win Rate	30% (Custom DAX measure)
Card	Total Deals	30 (COUNT records)
Card	Avg Deal Size	\$282.5M (AVERAGE Deal Value)
Bar Chart	Deal Count by Stage	Deal count by each pipeline stage
Column Chart	Monthly Deal Value	Total deal value by month (Jan–Apr)
Donut Chart	Stage Distribution	Percentage % deal theo stage
Table	Deal List	Deal Name, Customer, Value, Stage, Date, Rep
Slicer	Sales Rep Filter	Filter by Minh / Hoa / Tuan
Slicer	Stage Filter	Filter by pipeline stage



Overview Dashboard



Column Chart: Monthly Deal Value

7. COPILOT STUDIO - AI SALES ASSISTANT

The “Sales Assistant” AI agent was deployed on Copilot Studio to allow users to query pipeline data using natural language, without needing to open a separate dashboard or application.

7.1 Topic 1: Deals Closing Soon

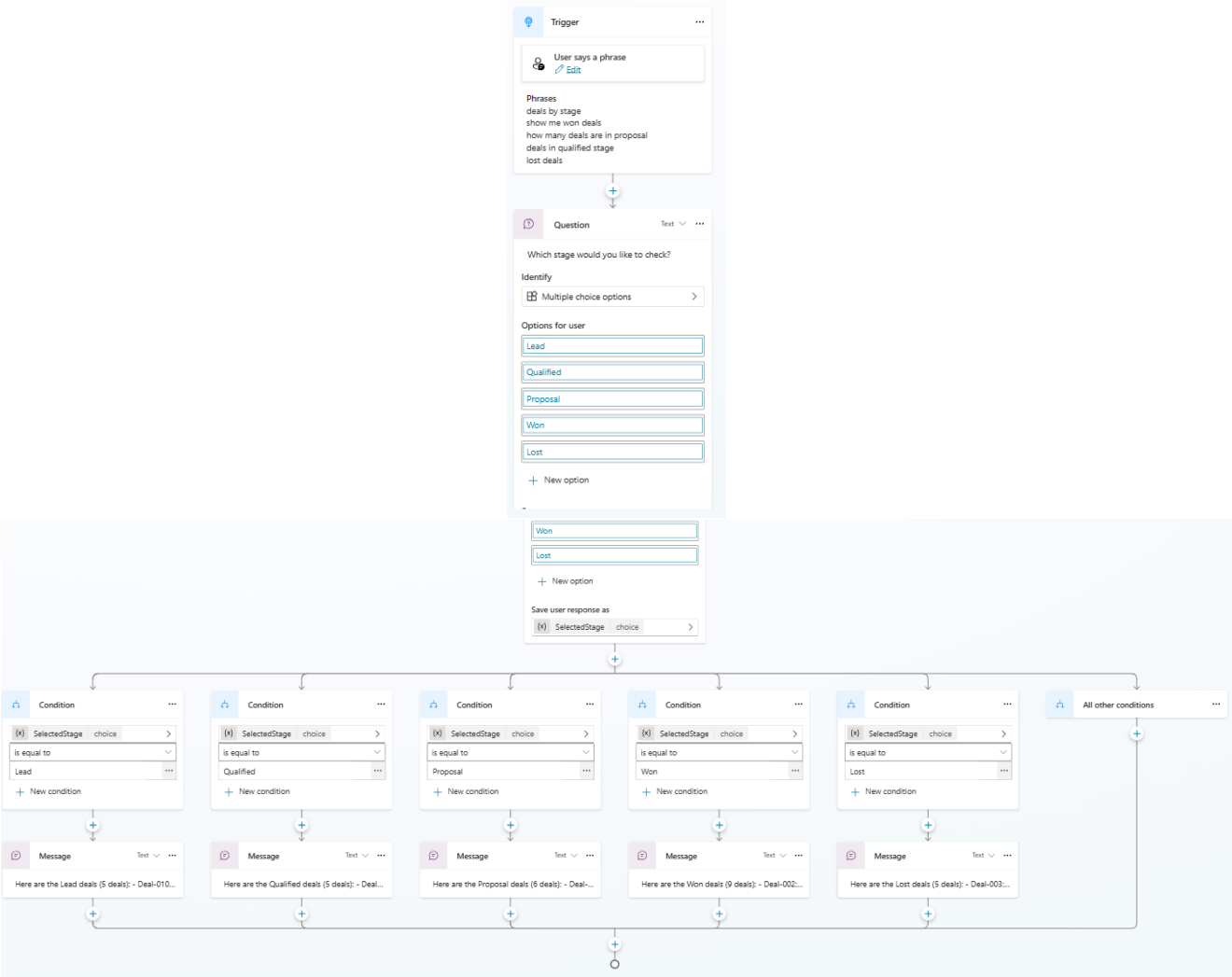
Property	Details
Trigger Phrases	deals closing soon / which deals are closing this week / upcoming deals
Response	List of the 4 soonest-closing deals with Name, Customer, Value, Rep, and Close Date

7.2 Topic 2: Total Revenue

Property	Details
Trigger Phrases	total revenue / how much revenue / what is our win rate / pipeline value
Response	Total Pipeline: \$8.475B Win Rate: 30% Total Deals: 30 Avg: \$282.5M + breakdown by Rep

7.3 Topic 3: Deals By Stage

Property	Details
Trigger Phrases	deals by stage / show me won deals / how many deals are in proposal
Interaction Flow	Bot asks the user which stage to view → 5 choices → Bot returns filtered results
Variable Used	SelectedStage (type: choice)
Implementation	Condition branching with 5 separate message nodes for each stage



Topic 3: Condition Branching

Test your agent

Hello, I'm Sales Assistant. How can I help?

5 minutes ago

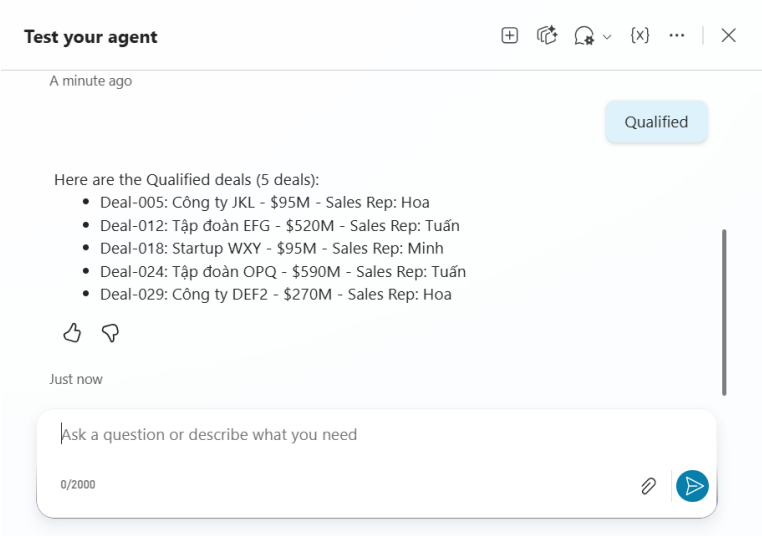
deals by stage

Which stage would you like to check?

A minute ago

Ask a question or describe what you need

0/2000



Demo Conversation Screenshot

7.4 Test Results

Test Case	Result
Topic 1: Returns correct list of soon-to-close deals	✓ PASS
Topic 2: Returns all KPI metrics correctly	✓ PASS
Topic 3: Bot correctly asks for stage confirmation	✓ PASS
Topic 3: Returns correctly filtered results by stage	✓ PASS

APPENDIX. DETAILED DATA

A. Deal Distribution by Sales Rep

Sales Rep	Deal Count	Percentage
Minh	10	33%
Hoa	10	33%
Tuan	10	33%

B. Data Source Information

- Dataverse Table: cre2e_deals - 30 records, 7 columns
- Imported from CSV/Excel, synthetic data Jan–Apr 2025
- 3 sales reps (Minh, Hoa, Tuan) - 10 deals each
- Stage distribution: 5 Lead, 5 Qualified, 6 Proposal, 9 Won, 5 Lost
- URL Dataverse: org16d99bb6.crm5.dynamics.com